

Name : _____

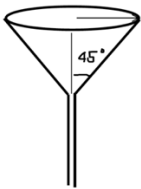
ChE 311 Midterm Exam II, Spring 2019
Closed book and closed notes. Only provided tables are allowed.
50 minutes, 100 points

1. (5 points) Which of the following statements is/are true?

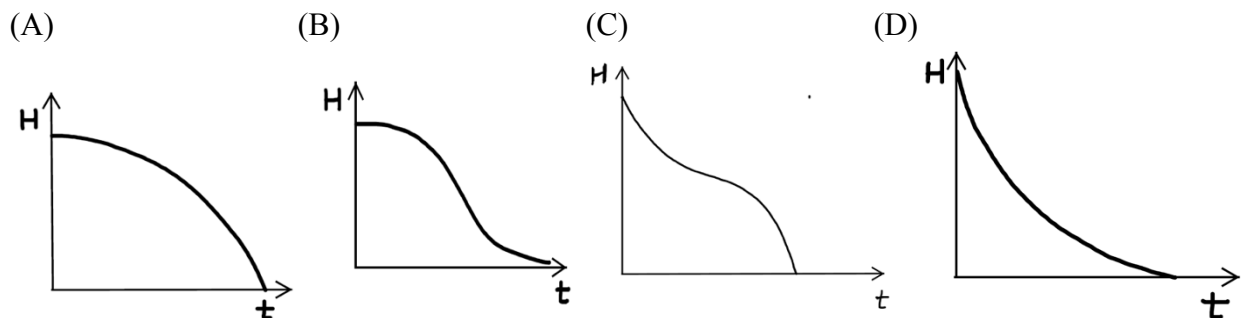
- (A) Eddy viscosity of Boussinesq is a property of the fluid.
- (B) Eddy viscosity for a flow system is constant everywhere.
- (C) Eddy viscosity of Boussinesq gives a first-order approximation for Reynolds Stress.
- (D) None of above.

Answer : _____

2. (5 points) Which of the following curve represents the draining curve of a conical tank?



Answer: _____



3. (5 points) Estimate the order of the Reynolds number of a swimmer who swims a kilometer in 20 minutes. Choose the closest answer.

- (A) 0.01
- (B) 1
- (C) 100
- (D) 100,000

Answer: _____

4. (5 points) For flow in a circular tube, the coordinate was set up as z was the flow direction and r was the radial direction. Please write down the $\bar{\tau}_{rz}^{(t)}$ by its definition in terms of velocity fluctuation?

5. (5 points) For above flow in a circular tube, please write down $\bar{\tau}_{rz}^{(t)}$ in terms of wall shear stress (τ_0) and the radius of the tube (R).

6. (30 points) **Turbulent flow in a circular tube.**

Please determine the ratio of $\mu^{(t)}/\mu$ at $y = R/3$ for water flowing at a steady rate in a long, smooth, circular tube under the following conditions.

Tube radius, $R = 10$ cm

Wall shear stress, $\tau_0 = 0.2$ Pa

Density of the fluid, $\rho = 1000$ kg/m³

Viscosity of the fluid, $\mu = 1$ cp

For viscous sublayer, $\frac{\bar{v}_x}{v_*} = \frac{yv_*}{\nu} \left[1 - \frac{1}{2} \left(\frac{\nu}{v_* B} \right) \left(\frac{yv_*}{\nu} \right) - \frac{1}{4} \left(\frac{yv_*}{14.5\nu} \right)^3 + \dots \right] \quad 0 < \frac{yv_*}{\nu} < 5$

For inertial sublayer, $\frac{\bar{v}_x}{v_*} = 2.5 \ln \left(\frac{yv_*}{\nu} \right) + 5.5 \quad \frac{yv_*}{\nu} > 30$

7. **(25 points) Using a conical funnel to measure the viscosity of a fluid.**

The dimensions of the conical funnel are shown in the figure. The initial liquid level is at $Z_1 = 5\text{ cm}$. The efflux time for completely draining the fluid (t_{efflux}) is 100s. $R_0 = Z_0 = 8\text{ cm}$. The pipe diameter (D) is 2mm. Pipe length (L) is 10 cm. Density of the fluid (ρ) has been measure, which is 1500 kg/m^3 . Take $g = 10\text{ m/s}^2$. Please decide the viscosity of the fluid.

